Lecture 3
The Business Investment Decision

- Understand the key principles of business investment decisions.
- Learn to evaluate investment opportunities using financial metrics.
- Explore the impact of risk.

What is Capital Budgeting? LT 12-2

Capital Budgeting:

- represents a long-term investment decision
- involves the planning of expenditures for a project with a life of many years
- usually requires a large initial cash outflow with the expectation of future cash inflows
- uses present value analysis

Introduction

- Concern:
  - Uncertainty (risk) about annual costs and inflows
  - Change in environment (external factors), e.g. technology, economic conditions, industry competitiveness
- How to incorporate risk into the decision.

Executive consideration

- Steps taken in the decision-making process
  - Search for and discover of investment opportunities
  - Collection of data
    - Engineering data
    - Marketing surveys
    - Likelihood of various event occurrence
  - Evaluation and decision making
  - Reevaluation and adjustment

See Figure 12-1

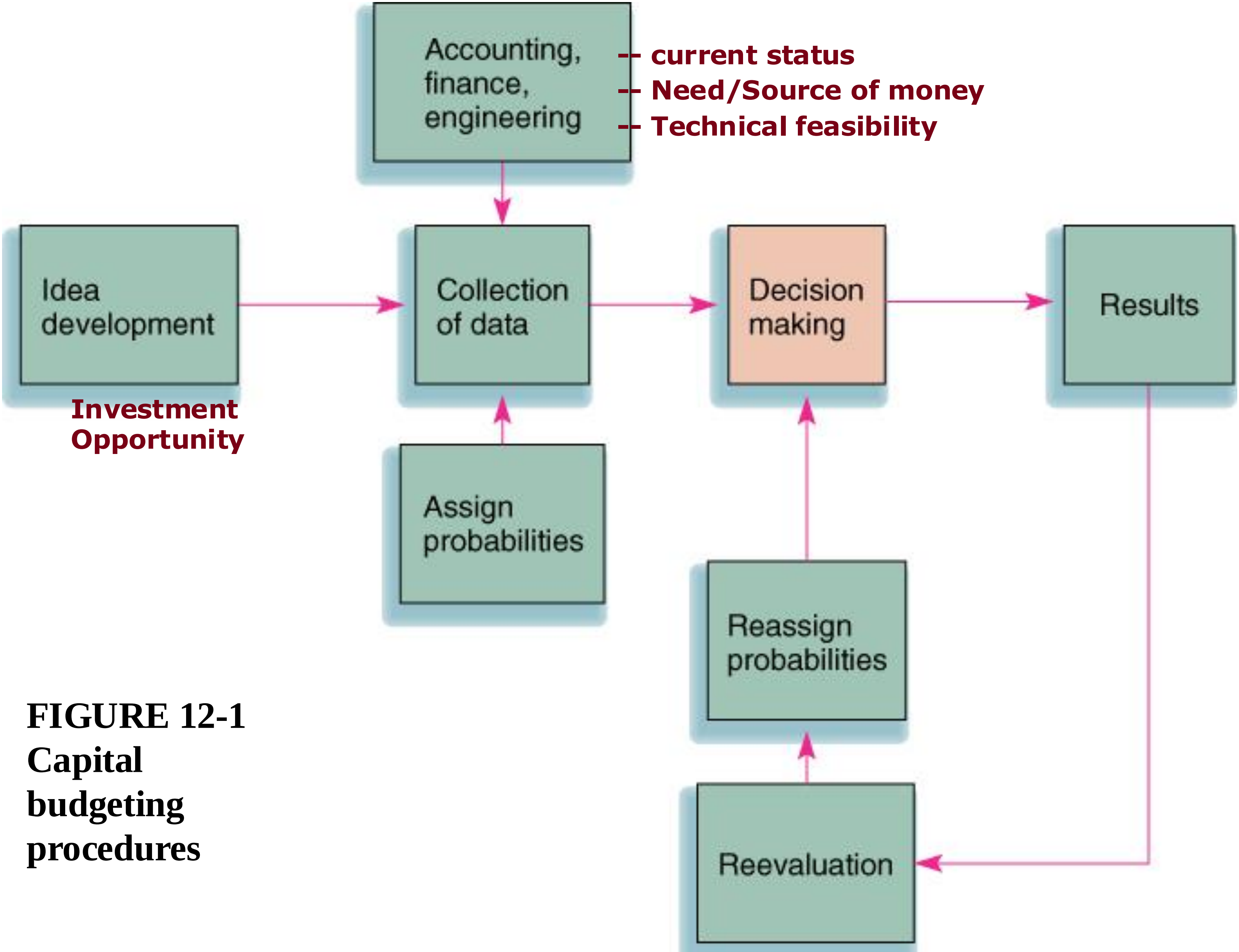


FIGURE 12-1
Capital budgeting procedures

Investment decision methods

Basic Methods for Investment Decision

- Payback Period
- Net Present Value (NPV)
- Profitability Index (PI)
- Internal Rate of Return (IRR)

Payback Method

– computes the amount of time required to recoup the initial investment

– a "**cutoff period**" is established

Advantages:

- easy to use ("quick and dirty" approach)
- emphasizes liquidity
  - In US, max. time horizon = 3-5 yrs..
  - A rapid payback may be important to those in industries with rapid technological change

See example

TABLE 12-3
Investment alternatives

	Cash Inflows (of $10,000 investment)	
Year	Investment A	Investment B
1	$5,000	$1,500
2	5,000	2,000
3	2,000	2,500
4		5,000
5		5,000

- Payback period for investment A is 2 years
- Payback period for investment B is 3.8 years
  - Interpolate of 4000 : 5000 * (year 4-3 = 1) → 0.8 * 1
- *Total inflows of A = $12,000 and B = $16,000, Which one is better in terms of payback period?*

Discuss disadvantage

Payback period (ii)

Disadvantages:

(1) ignores inflows after the cutoff period and

(2) fails to consider the time value of money

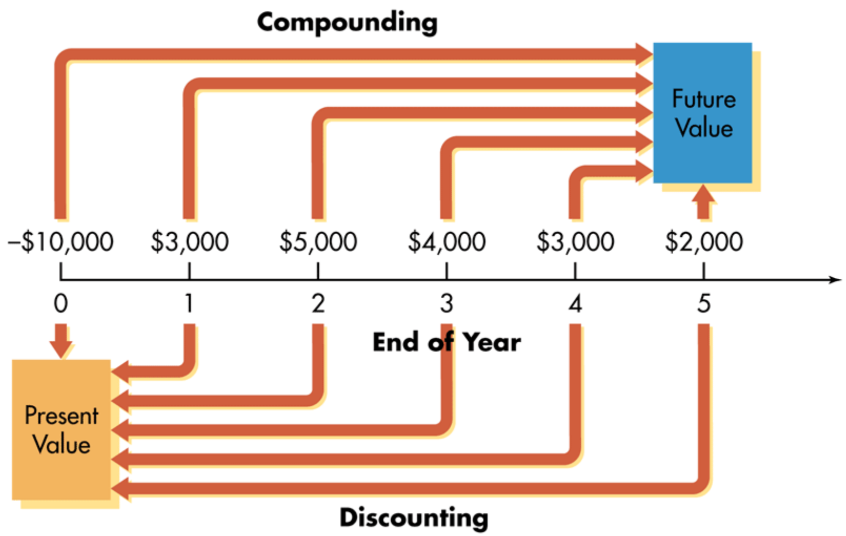
- Compare the project with inflows of $1000 and $9000 and another one with inflows of $9000 and $1000?

Net Present Value

Net Present Value (NPV):

- the present value of the cash inflows minus the present value of the cash outflows
- the cash inflows are discounted back over the life of the investment
- the basic discount rate is usually the firm's cost of capital
  - Average of debt and common equity (stock)

Compounding and Discounting of cashflow



Discounting
- Finding Present Value (PV)

$$PV = \frac{FV_n}{(1+i)^n} = FV_n\left(\frac{1}{1+i}\right)^n$$

NPV 10% discount rate

ear	Investment A	
1	$5,000 × 0.909 =	$ 4,545
2	5,000 × 0.826 =	4,130
3	2,000 × 0.751 =	1,502
		$10,177
esent value of inflows		$10,177
esent value of outflows		−10,000
et present value		$177

Year	Investment B	
1	$1,500 × 0.909 =	$ 1,364
2	2,000 × 0.826 =	1,652
3	2,500 × 0.751 =	1,878
4	5,000 × 0.683 =	3,415
5	5,000 × 0.621 =	3,105
		$11,414
Present value of inflows		$11,414
Present value of outflows		−10,000
Net present value		$ 1,414

- By NPV, which investment is chosen, why?

Profitability Index (PI)

- The present value of a project's future cash flows divided by the initial investment

$$PI = \frac{PV\ of\ future\ cashflows}{Initial\ investment} = 1 + \frac{NPV}{Initial\ investment}$$

- PI measures how much investors get for each Dollar they invest in the project
- When NPV is positive:

  PI > 1
- When NPV is negative:

  PI < 1

Investment A, PI = 10177/10000 = 1.017

Investment B, PI = 11414/10000 = 1.141

- Decision rule = Invest, only when PI > 1
- PI is commonly referred to ***benefit-cost ratio***

Internal Rate of Return

Internal Rate of Return (IRR):

- represents *a yield* or *an interest rate on an investment*
- requires calculating the interest rate that equates the cash outflows (cost) with the cash inflows
- Or the interest rate that makes NPV = 0

$$\frac{CF_t}{(1+IRR)^t} = Outlay$$

E.g. For investment A:

$$-10{,}000 + \frac{5{,}000}{(1+IRR)^1} + \frac{5{,}000}{(1+IRR)^2} + \frac{2{,}000}{(1+IRR)^3} = 0$$

IRR: Trial and Error for investment A

Year	10%	
1	$5,000 X 0.909	$4,545
2	5000 X 0.826	4,130
3	2,000 X 0.751	1,502
		$10,177

Year	12%	
1	$5,000 X 0.893	$4,465
2	5000 X 0.797	3,985
3	2,000 X 0.712	1,424
		$9,874

- To be more accurate, the results can be interpolated
  - Distance of 10177 – 9874 = $303 [10% -> 12%]
  - Distance of 10177 – 10000 = $177
    - (177 / 303) * 2 = 1.17%
    - So, the answer is 10% + 1.17 = 11.17% IRR
- For investment B, the same process give IRR = 14.33%, ***which investment is better, then?***

Selection strategy

- For IRR and NPV, to be acceptable, the profitability must equal or exceed the cost of capital
- If investments (projects) are *mutually exclusive*, then only one of which will be picked.
- Example

Alternatives	IRR	NPV
Bangkok	15%	$300
Beijing	12	200
Mexico City	11	100
Cost of capital	10	-
Singapore	9	-90

  - If the investment is mutually exclusive, which project will be selected?
  - If not mutually exclusive, which projects will be selected?
- What about the investments in Table 12-3, if cost of capital is 12%

Accept/Reject Decision

Payback Method (PB):

- if PB period < cutoff period, <u>accept</u> the project
- if PB period > cutoff period, <u>reject</u> the project

Net Present Value (NPV):

- if NPV > 0, <u>accept</u> the project
- if NPV < 0, <u>reject</u> the project

Internal Rate of Return (IRR):

- if IRR > cost of capital, <u>accept</u> the project
- if IRR < cost of capital, <u>reject</u> the project

Ranking conflict between NPV and IRR

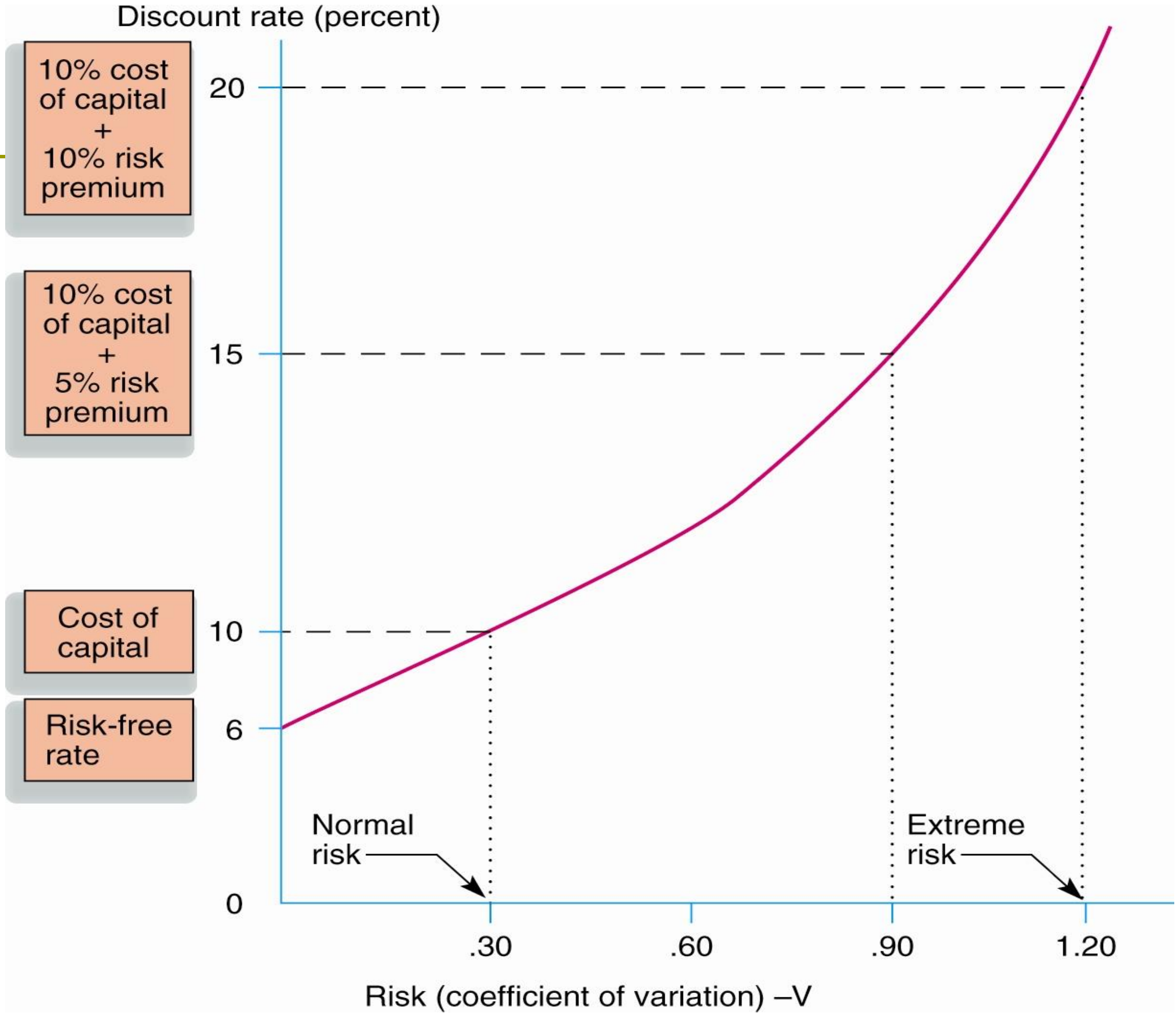
r	10%							
Year	0	1	2	3	4	NPV	PI	IRR
Project A	-100	50	50	50	50	$53.18	$1.53	34.90%
Project B	-400	170	170	170	170	$126.25	$1.32	25.21%

- Project A has a much smaller outlay than project B, although they have similar cash flow pattern
- However, size of project B is larger than size of project A
- Would you rather have a small project with higher IRR or a large project with lower IRR?

Risk-Adjusted Discount Rate

- Different capital expenditure proposals with different risk levels would require different discount rates
  - Project with normal amount of risk discounted at the cost of capital
  - Project with greater than normal risk discounted at higher rate (as shown in the following slide)
- Risk is assumed to be measured by the coefficient of variation (V)

Relationship of Risk to Discount Rate



- **This is an example of being increasingly risk-averse at higher levels of risk and potential return**

Increasing Risk over Time

- Accurate forecasting becomes more obscure farther out in time
- Unexpected events:
  - Create a higher standard deviation in cash flows
  - Increase risk associated with long-lived projects
- Use of progressively higher discount rates to compensate for risk tends to penalize later cash flows more than earlier ones

Qualitative Measures

- Setting up of risk classes based on qualitative considerations
- This again equates the discount rate to the perceived risk
- Example of such risk classes illustrated in the following slide

Risk Categories and Discount rates

	Discount Rate
Low or no risk (repair to old machinery)	6%
Moderate risk (new equipment)	8
Normal risk (addition to normal product line)	10
Risky (new product in related market)	12
High risk (completely new market)	16
Highest risk (new product in foreign market)	20

Capital Budgeting Analysis

- **This following table shows Investment B as a preferred investment based on NPV calculation without considering the risk factor**

Table 13–4

Year	Investment A (10% discount rate)		Year	Investment B (10% discount rate)	
1	$5,000 × 0.909 =	$ 4,545	1	$1,500 × 0.909 =	$ 1,364
2	5,000 × 0.826 =	4,130	2	2,000 × 0.826 =	1,652
3	2,000 × 0.751 =	1,502	3	2,500 × 0.751 =	1,878
		$10,177	4	5,000 × 0.683 =	3,415
			5	5,000 × 0.621 =	3,105
					$11,414
Present value of inflows		$10,177	Present value of inflows		$11,414
Investment..................................		−10,000	Investment..................................		−10,000
Net present value		$ 177	Net present value		$ 1,414

Capital Budgeting Decision Adjusted for Risk: Example

- **Investment A** calls for an addition to normal product line and is assigned discount rate of 10%
- **Investment B** represents a new product in foreign market and must carry 20% discount rate to adjust for a large risk component

Capital Budgeting Decision Adjusted for Risk: Example (cont'd)

Year	Investment A (10% discount rate)		Year	Investment B (20% discount rate)	
1	$5,000 × 0.909 =	$ 4,545	1	$1,500 × 0.883 =	$ 1,250
2	5,000 × 0.826 =	4,130	2	2,000 × 0.694 =	1,388
3	2,000 × 0.751 =	1,502	3	2,500 × 0.579 =	1,448
		$10,177	4	5,000 × 0.482 =	2,410
			5	5,000 × 0.402 =	2,010
					$ 8,506

Present value of inflows	$10,177	Present value of inflows	$ 8,506
Investment..................................	−10,000	Investment..................................	−10,000
Net present value	$ 177	Net present value	$ (1,494)

- **Investment A is the only acceptable alternative after adjusting the risk factor**

Combining cash flow analysis and selection strategy

The rule of depreciation

- calculate a depreciation schedule using the appropriate MACRS class
  - MACRS = Modified accelerated cost recovery system
  - Some referred as ADR = asset depreciation range
  - Asset can be written off more rapidly than the midpoint of their ADR
  - If midpoint of an asset's useful life = 4 years, then it might be written off over 3 years. (Table 12-8)
- The rate of depreciation (in US) is developed with the use of the half-year convention (treats as if it were put in service in midyear)
  - Refer to Table 12-9
- Table 12-10 shows the depreciation schedule for a $50,000 asset you purchase, if it falls in 5-yr MACRS category.

TABLE 12-8
Categories for depreciation write-off

PPT 12-8

Class	
3-year MACRS	All property with ADR midpoints of four years or less. Autos and light trucks are excluded from this category.
5-year MACRS	Property with ADR midpoints of more than 4, but less than 10 years. Key assets in this category include automobiles, light trucks, and technological equipment such as computers and research-related properties.
7-year MACRS	Property with ADR midpoints of 10 years or more, but less than 16 years. Most types of manufacturing equipment would fall into this category, as would office furniture and fixtures.
10-year MACRS	Property with ADR midpoints of 16 years or more, but less than 20 years. Petroleum refining products, railroad tank cars, and manufactured homes fall into this group.
15-year MACRS	Property with ADR midpoints of 20 years or more, but less than 25 years. Land improvement, pipeline distribution, telephone distribution, and sewage treatment plants all belong in this category.
20-year MACRS	Property with ADR midpoints of 25 years or more (with the exception of real estate, which is treated separately). Key investments in this category include electric and gas utility property and sewer pipes.
27.5-year MACRS	Residential rental property if 80% or more of the gross rental income is from nontransient dwelling units (e.g., an apartment building); low-income housing.
31.5-year MACRS	Nonresidential real property is real property that has no ADR class life or whose class life is 27.5 years or more.
39-year MACRS	Nonresidential real property placed in service after May 12, 1993.

TABLE 12-9 Depreciation percentages (expressed in decimals)

Depreciation Year	3-Year MACRS	5-Year MACRS	7-Year MACRS	10-Year MACRS	15-Year MACRS	20-Year MACRS
1	.333	.200	.143	.100	.050	.038
2	.445	.320	.245	.180	.095	.072
3	.148	.192	.175	.144	.086	.067
4	.074	.115	.125	.115	.077	.062
5		.115	.089	.092	.069	.057
6		.058	.089	.074	.062	.053
7			.089	.066	.059	.045
8			.045	.066	.059	.045
9				.065	.059	.045
10.......				.065	.059	.045
11.......				.033	.059	.045
12.......					.059	.045
13.......					.059	.045
14.......					.059	.045
15.......					.059	.045
16.......					.030	.045
17.......						.045
18.......						.045
19.......						.045
20.......						.045
21.......						.017
	1.000	1.000	1.000	1.000	1.000	1.000

Depreciation write-off rate (Thailand)

- According to GFMIS (กรมบัญชีกลาง)

Asset Category	Depreciation Rate	Useful Life (Years)
Buildings	5%	20
Machinery and Equipment	10%	10
Vehicles	20%	5
Office Equipment	20%	5
Computer Equipment	33.33%	3

Investment decision with more practical

To make the investment decision in a more realistic business case:

- Calculate the depreciation schedule
- Figure earnings, expenses and cash flow (CF)
- Discount the cash flows back to the present to determine whether the machine should be purchased (only if NPV > 0)

Sample Case of Investment decision

Example: a $50,000 Investing decision:

Given the depreciation schedule as in Table 12-10 (5-yrs MACRS), if the project will generate income of $18,500 for the first 3 yrs, and $12,000 for the last 3 yrs, tax rate of 35%, and a cost of capital of 10%, should we invest this project?

TABLE 12-10
Depreciation schedule

(1) Year	(2) Depreciation Base	(3) Percentage Depreciation (Table 12–9)	(4) Annual Depreciation
1	$50,000	.200	$10,000
2	50,000	.320	16,000
3	50,000	.192	9,600
4	50,000	.115	5,750
5	50,000	.115	5,750
6	50,000	.058	2,900
		Total Depreciation	$50,000

TABLE 12-11
Cash flow related to the purchase of machinery

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Earnings before depreciation and taxes (EBDT)	$18,500	$18,500	$18,500	$12,000	$12,000	$12,000
Depreciation (from Table 12–10)	10,000	16,000	9,600	5,750	5,750	2,900
Earnings before taxes	8,500	2,500	8,900	6,250	6,250	9,100
Taxes (35%)	2,975	875	3,115	2,188	2,188	3,185
Earnings after taxes	5,525	1,625	5,785	4,062	4,062	5,915
+ Depreciation	10,000	16,000	9,600	5,750	5,750	2,900
Cash flow	$15,525	$17,625	$15,385	$ 9,812	$ 9,812	$ 8,815

TABLE 12-12
Net present value analysis

Year	Cash Flow (inflows)	Present Value Factor (10%)	Present Value
1	$15,525	.909	$14,112
2	17,625	.826	14,558
3	15,385	.751	11,554
4	9,812	.683	6,702
5	9,812	.621	6,093
6	8,815	.564	4,972
			$57,991

Present value of inflows	$57,991
Present value of outflows (cost)	50,000
Net present value	$ 7,991

All the cash flows in Table 12-11 are discounted back to the present with the cost of capital at 10%

Should we buy the machine?

Topic review

- 3 main methods for capital budgeting
  - Payback period, IRR and NPV
- Each, having different criteria for deciding accept or reject the projects.
- To be realistic, the consideration of capital budgeting should consider the effect of depreciation and tax policies to the net cash flow
- Further effective capital budgeting can be achieved by adjusting the discount rate according to the perceived risk.

Further readings

- Capital budgeting concepts http://www.edupristine.com/blog/capital-budgeting-techniques
- Capital budgeting and risk analysis http://educ.jmu.edu/~drakepp/principles/module6/cbrisk.pdf
- Cost of capital http://www.accountingtools.com/cost-of-capital